

# Supplementary Material for “A Semantic Mutation Metric for Metamorphic-Relation Adequacy in Scientific Computing Programs”

MENG LI, School of Computing, University of South China, China, Hunan Engineering Research Center of Software Evaluation and Testing for Intellectual Equipment, China, and CNNC Key Laboratory on High Trusted Computing, China

XIAOHUA YANG, School of Computing, University of South China, China

JIE LIU, School of Computing, University of South China, China

SHIYU YAN, School of Computing, University of South China, China

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation*.

Additional Key Words and Phrases: metamorphic testing, mutation testing, semantic mutation testing, metamorphic relation adequacy, metamorphic relation assessment, scientific computing

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## A Notation and Operator Catalogue

### A.1 Complete notation table

*Experimental objects.*

- PUT set  $I = \{A1, A2, A3, B1, B2, B3, C1, C2, C3, D1, D2, D3\}$ ,  $|I| = 12$ .
- PUT:  $S_i$  ( $i \in I$ ).
- Class mapping:  $\text{cls} : I \rightarrow \{A, B, C, D\}$  (open, extensible).
- Valid input distribution:  $D_S$ , sampling  $X_{K_{\text{eq}}} \sim D_S$ .

*Metamorphic testing side.*

- Metamorphic-relation label set:  $\mathcal{R} = \{\text{Cons.}, \text{Mono.}, \text{Conv.}, \text{Traj.}, \text{P-ord.}\}$ ,  $k \in \{1, \dots, 5\}$  (open, extensible).
- Metamorphic Relation:  $\text{MR}_{i,k}$  (from the shared infrastructure);  $mr = (r, R) \in \text{MR}_{i,k}$ .
- Automated Verification Pipeline (AVP):  $\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \rightarrow \{\text{pass}, \text{fail}\}$ ;  $\text{AVP}(s, mr, \epsilon_{\text{AVP}}^k)$  invokes the verification method per relation index  $k$ .

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Authors' Contact Information: Meng Li, [mlemon@usc.edu.cn](mailto:mlemon@usc.edu.cn), School of Computing, University of South China, Hengyang, China and Hunan Engineering Research Center of Software Evaluation and Testing for Intellectual Equipment, Hengyang, China and CNNC Key Laboratory on High Trusted Computing, Hengyang, China; Xiaohua Yang, [xiaohua1963@foxmail.com](mailto:xiaohua1963@foxmail.com), School of Computing, University of South China, Hengyang, China; Jie Liu, [jjeliu@usc.edu.cn](mailto:jjeliu@usc.edu.cn), School of Computing, University of South China, Hengyang, China; Shiyu Yan, [yanshiyu@usc.edu.cn](mailto:yanshiyu@usc.edu.cn), School of Computing, University of South China, Hengyang, China.

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*Mutation testing side.*

- Mutation operator family:  $MUT = \{\text{mut}_1, \dots, \text{mut}_5\}$ ,  $j \in \{1, \dots, 5\}$  (open, extensible);  $\text{mut}_1 = \text{mut}_{\text{inv}}$ ,  $\text{mut}_2 = \text{mut}_{\text{mono}}$ ,  $\text{mut}_3 = \text{mut}_{\text{conv}}$ ,  $\text{mut}_4 = \text{mut}_{\text{dyn}}$ ,  $\text{mut}_5 = \text{mut}_{\text{cmp}}$ .
- Signature:  $\text{mut}_j : \text{Programs} \rightarrow 2^{\text{Programs}}$ .
- Mutant:  $s' \in \text{mut}_j(S_i)$ .
- Alignment:  $\text{align}(j) = j$ .

*Three-state decomposition (fixed).*

$$\text{mut}_j(S_i) = \text{equiv}_{i,k,j} \cup \text{killed}_{i,k,j} \cup \text{survive}_{i,k,j},$$

with the three sets disjoint, mutually exclusive, and exhaustive.

- *equiv determination* (dual conditions): (E1) AVP-coherent:  $\forall mr \in MR_{i,k} : \text{AVP}(S_i, mr) = \text{AVP}(s', mr)$ ; (E2) Output-equiv:  $\forall x \in X_{K_{\text{eq}}} \sim D_S : \|S_i(x) - s'(x)\| \leq \epsilon_{\text{eq}}$ .
- *killed determination*:  $\text{killed}(s', MR_{i,k}) \Leftrightarrow \exists mr \in MR_{i,k} : \text{AVP}(S_i, mr) = \text{pass} \wedge \text{AVP}(s', mr) = \text{fail}$ .

*Metric (fixed classical structure).*

$$\begin{aligned} \text{SMS}_{i,k,j} &:= \frac{|\text{killed}_{i,k,j}|}{|\text{mut}_j(S_i)| - |\text{equiv}_{i,k,j}|} \\ &= \frac{|\text{killed}_{i,k,j}|}{|\text{killed}_{i,k,j}| + |\text{survive}_{i,k,j}|} \in [0, 1]. \end{aligned}$$

*LRCA engineering attribution layer (descriptive, not in SMS).*

- Likely root-cause inventory:  $C = \{C_1, \dots, C_5\}$  (open, extensible). C1 True semantic failure; C2 Tolerance perturbation; C3 OOD; C4 Statistical-assumption violation; C5 Mutator artefact.
- LRCA output: each  $s' \in \text{killed}$  annotated with  $\text{root\_cause}(s') \in C$ .
- Descriptive quantities:  $C1\_share_{i,k,j}$ ,  $\text{suspect\_share}_{i,k,j}$ .

*Slicing and cross-class.*

- Slicing: aligned  $j = k$ ; cross  $j \neq k$ .
- Cross-class:  $\Delta \text{SMS}_c$ ,  $c \in \{A, B, C, D\}$ ;  $\text{CV}(\Delta \text{SMS}) := \text{std}(\Delta \text{SMS}_c) / |\text{mean}(\Delta \text{SMS}_c)|$ .

*Tolerance and sampling.*

- Tolerance:  $\epsilon_{\text{AVP}}^k$  (relation-dependent),  $\epsilon_{\text{eq}}$  (equivalence output tolerance).
- Sampling:  $K_{\text{eq}} = 1000$  (E2 sample size),  $N = 20$  (statistical replicates).

The notation skeleton (11 core concepts + three-state decomposition + SMS formula + AVP interface) is **fixed**; the *content* of  $MUT$ ,  $\mathcal{R}$ ,  $C$ , and  $\text{cls}$  is open to extension.

## A.2 Root-Cause Decision Tree and Diagnostic Protocol

**Five likely root causes** (open):

Table 1. Root-cause decision tree and three-layer diagnostic protocol.

| Code | Meaning                                |
|------|----------------------------------------|
| C1   | True semantic failure (what SMS seeks) |
| C2   | Numerical tolerance perturbation       |
| C3   | Out-of-distribution (OOD)              |
| C4   | Statistical-assumption violation       |
| C5   | Mutator artefact                       |

**Three-layer diagnosis with L0 artefact pre-scan:**

- **L0 artefact pre-scan.** Double-blind review following the main manuscript’s mutant-pool prescreen and equivalence-detection protocol: dual-LLM cross-source review plus 20% manual sampling.
- **L1 tolerance robustness.**  $N = 20$  replicates; fail ratio  $\geq 0.80$  considered stable.
- **L2 OOD triage.** For C/D classes, distinguish valid  $D_S^{\text{valid}}$  from OOD.
- **L3 statistical baseline.** For Wilcoxon/DTW, IID/stationarity pre-check on PUT samples.

**Decision tree:**

For each  $s' \in \text{killed}_{i,k,j}$ :

- **L1.** fail ratio  $< 0.80 \Rightarrow \text{C2}$ ; otherwise  $\Rightarrow \text{L2}$ .
- **L2** (C/D classes). fail only in OOD  $\Rightarrow \text{C3}$ ; otherwise  $\Rightarrow \text{L3}$ .
- **L3** (B/D classes + Wilcoxon/DTW). assumption violation  $\Rightarrow \text{C4}$ ; otherwise  $\Rightarrow \text{artefact recheck}$ .
- **Artefact recheck.** artefact evidence  $\Rightarrow \text{C5}$ ; otherwise  $\Rightarrow \text{C1}$ .

**Multi-label priority:**  $\text{C5} > \text{C4} > \text{C3} > \text{C2} > \text{C1}$ .

**LRCA threshold calibration (9-grid scan; lrca\_calibration.json).** Best ood\_band=0.02 with any tolerance\_multiplier (3.0 / 10.0 / 30.0) gives mean\_C1\_share 0.200 / H4 12/60 (20.0%); ood\_band=0.05 (default) or 0.10 gives 0.164 / 10/60 (16.7%). tolerance\_multiplier has zero impact (L1 tolerance rarely triggered); ood\_band is the only discriminator. Calibration ceiling 12/60 = 20% remains far below 80%, H4 unattainable on this dataset, not a calibration issue (inherent SNR of LLM-mutant pools).

**A.3 Verification Protocol and Equivalence-Judgement Trade-Off**

$$\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \rightarrow \{\text{pass}, \text{fail}\}.$$

Per-relation verification: Cons. tolerance equality  $|\text{LHS} - \text{RHS}| \leq \varepsilon$ ; Mono. / P – ord. Wilcoxon signed-rank  $\alpha = 0.05$ ; Conv. convergence-order + asymptotic residual ratio; Traj. DTW distance threshold  $\varepsilon_{\text{DTW}}$ . AVP version = upstream infrastructure commit hash; our package embeds AVP source.

**Three-candidate equivalence trade-off.**

Table 2. Verification protocol and equivalence-judgement trade-off.

| Determination                    | False-positive                       | False-negative                             | SMS bias                     |
|----------------------------------|--------------------------------------|--------------------------------------------|------------------------------|
| E1 alone                         | numerical coincidence + AVP-coherent | E1 false $\rightarrow$ non-equiv           | Low (smaller denominator)    |
| E2 alone                         | output match, MR differs (rare)      | K_eq miss, full-space equal                | High                         |
| <b>E1 <math>\wedge</math> E2</b> | both mis-pass simultaneously         | E1 $\vee$ E2 false $\rightarrow$ non-equiv | Slightly high (conservative) |

E1 alone is fooled by insufficient AVP coverage; E2 alone by numerical coincidence on  $K_{eq}$ . E1  $\wedge$  E2 is the conservative complete instantiation; under  $L_{equiv}$  it reduces almost-everywhere to classical bitwise equivalence (Lemma G.1). Counter-examples: E2 passes but E1 does not (rare; mutant coincides at  $K_{eq}$  but deviates at MR trigger points  $\rightarrow$  non-equivalent); E1 passes but E2 does not (common; consistent within MR but numerical drift  $> \epsilon_{eq} \rightarrow$  non-equivalent).

## B Experimental Subjects and Mutation-Operator Specialisations

### B.1 Program Selection Rationale

**(a) Library-stack coverage.** numpy 2.4.4 (A1-B3 linear algebra / arrays), scipy 1.17.1 (A1 integrate, A2 linalg, B2 stats), scikit-learn 1.8.0 (C1-D3 surrogate / ML), Python scientific-computing’s de facto foundation stack (PyPI top-50, 2026-04). Uncovered: GPU / distributed (JAX, CuPy, dask), domain-specific (BioPython, Astropy, RDKit). Left to future work under the representativeness threat.

**(b) Mathematical-structure coverage.** ODE/PDE (A1, A3), direct linear algebra (A2), Bayesian analytic + MCMC + Monte Carlo (B1-B3), kernel + orthogonal polynomial + NN surrogate (C1-C3), convex optimisation + backprop + max-likelihood (D1-D3), covers 8 of 12 chapters in Numerical Recipes [2]. Uncovered: advanced PDE solvers (FEM, FV, spectral); FFT; interior-point/trust-region optimisation; symbolic/CAS. Left to future work.

**(c) Comparison with benchmarks.** DeepCrime (Humbatova et al. 2021): semantic class-D topical overlap on ML kernels. Defects4J (Just et al. 2014): no overlap, only mutation-as-fault-proxy reference. mutmut / cosmic-ray demos: general Python; the semantic operators extend the scientific-computing focus.

**(d) Scale vs representativeness.** Each PUT 50-400 LOC; signature standardised to `program(x: float)  $\rightarrow$  float`. Substantively smaller than industrial code (typically 1-10 KLOC).

**Limitation.** The `float  $\rightarrow$  float` signature is a substantive constraint, not engineering trade-off. Industrial inputs are typically high-dimensional (CFD grids, MD particles, FEM matrices); scalarised PUTs upper-bound mutant semantic complexity and may systematically under-estimate SMS and cross-class differences on industrial PUTs (declared in the main manuscript’s threats-to-validity section; future work validates on industrial-grade PUTs).

### B.2 Systematic vs incidental

Syntactic tools may *occasionally* hit the semantic-mutation criteria in the main manuscript (a) (cross-function-boundary) or (c) (algorithm class), but occasionality undermines two engineering functions of semantic mutation.

**(a) Deepening source-code understanding.** Designing a numerical-API substitution `np.linalg.det(M)  $\rightarrow$  np.sum(np.diag(M))` requires knowing the two are equivalent on diagonal matrices but not on general matrices

( $\det = \prod \text{eigvals}$  vs  $\text{sum}(\text{diag}) = \text{trace} = \sum \text{eigvals}$ ); syntactic tool AST traversal does not require this understanding even when similar replacements appear. **(b) Revealing deep faults.** Domain-semantic errors (physical-constant, unit conversion, boundary conditions, hyperparameter semantics, numerical-method order) are not AST-local; syntactic mutator design goals (operator typos, off-by-one, negation flips) have hit probability for domain errors much smaller than systematic semantic-mutator triggering, and lack repeatability. **Conclusion.** Syntactic tools occasionally producing mutants satisfying (a)(b)(c) is stochastic byproduct, not repeatable, not carrying engineering value. Systematic semantic mutation requires (a)(b)(c) to be design intent.

### B.3 Per-class operator specialisations

Table 3. Per-class operator specialisations.

| Operator                                          | Representative specialisations across PUTs                                                                                                                                                                                                                                                           |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{mut}_{\text{inv}}$ Conservation-breaking   | A1 Lorenz: add $\epsilon_{\text{drift}}$ to RHS (slow Hamiltonian drift); A2 LU: decomposition omits $k+1$ -th multiplier (breaks det conservation); B1 Beta-Bin: posterior omits normalisation; C1 GPR: covariance omits positive-definite diagonal term; D1 MLP: backprop omits one gradient term. |
| $\text{mut}_{\text{mono}}$ Monotonicity-breaking  | A3 FDM: $\Delta t$ coefficient occasionally negative; B2 MCMC: acceptance $\min(1, r) \rightarrow \min(0.95, r)$ ; C2 PCE: high-order coefficient sort inserts inversion; D2 SVM: decision-function sign flips near boundary.                                                                        |
| $\text{mut}_{\text{conv}}$ Convergence-breaking   | A1 Lorenz: RK4 $\rightarrow$ 1.5-order hybrid; A3 FDM: 2nd-order difference $\rightarrow$ 1st-order; B3 MC: doubling sample size does not $1/N$ -reduce variance; C3 NN-Surr: training-epoch truncation.                                                                                             |
| $\text{mut}_{\text{dyn}}$ Trajectory-distorting   | A1 Lorenz: state-vector $y/z$ swap; B2 MCMC: insert independent-sampling segment; C3 NN-Surr: training-target slow phase shift; D1 MLP: hidden-layer periodic-mask activation.                                                                                                                       |
| $\text{mut}_{\text{cmp}}$ Fidelity-order-breaking | A2 LU: partial pivoting degrades to no pivoting; C1 GPR: length-scale switches to coarse prior; C2 PCE: high-order term randomly retains low-order; D3 LR: regularisation occasionally large.                                                                                                        |

CE/OS/HP/TF/SI are historical internal campaign IDs for edit-mechanism families, retained for reproducibility; they are not reader-facing semantic classes. **Per-class HP / OS / TF substitution rules** (illustrative). **HP** on class a: tolerance / max\_iter; on class c: GPR noise\_level / length\_scale; on class d: MLP hidden\_dim / dropout. **OS** on class a: numerical-linalg API swaps ( $\det \leftrightarrow \text{sum}(\text{diag})$ ); on class b: probability-distribution sampling swaps; on class c:

surrogate-class swaps (GPR  $\leftrightarrow$  RBF  $\leftrightarrow$  NN). **TF** on class a: integration-order changes (RK4  $\rightarrow$  Euler); on class b: MC estimator changes.

**Operator-level cross-table with cosmic-ray defaults.** The 12 cosmic-ray default operators are AST-local: NumberReplacer (Num/Constant, CE partial,  $\Delta$  literal values only); ReplaceArithmeticOperator (BinOp); ReplaceComparisonOperator (Compare); ReplaceLogicalOperator (BoolOp); ReplaceUnaryOperator (UnaryOp); ReplaceTrueFalse (NameConstant, CE keyword,  $\Delta$  Bool literal); BreakContinueReplacer (Break/Continue); RemoveDecorator; RemoveExceptionHandler; ZeroIterationForLoop (For); ReplaceIfBlock (If); MutateSubscript. None correspond to OS / HP / TF / SI:

Table 4. Per-class operator specialisations.

| Semantic class                      | Tool support | Coverage                                                  |
|-------------------------------------|--------------|-----------------------------------------------------------|
| <b>OS</b> API replacement           | None         | $\Delta$ 88.33% disjoint (incidental low-complexity hits) |
| <b>HP</b>                           | None         | $\times$ tool inexpressible (0/72)                        |
| <b>TF</b> numerical-method order    | None         | $\times$ tool inexpressible (0/54)                        |
| <b>SI / CF</b> structural injection | None         | $\times$ tool inexpressible (0/33)                        |

**Operator-level conclusion.** All 12 default classes remain AST-local (BinOp / Compare / BoolOp / UnaryOp / NameConstant / Subscript / If / For / Break / Continue / decorator / except). Categorically, no entry recognises sklearn / scipy hyperparameter semantics (HP), numerical-method order (TF), or control-flow intent (SI / CF). For OS, low-complexity sub-expressions can be incidentally hit by BinOp; empirical 11.67% (the main manuscript’s AST-overlap audit) is consistent with 88.33% AST-disjointness lower bound.

#### B.4 Syntactic-Mutation Pilot Study

Before the 12-PUT generalisation in the main manuscript’s AST-overlap audit, a single-PUT lightweight empirical used scripts/run\_cosmic\_ray\_a1.sh on a1 (file 934 B, AST-bound). Outcome: mutants\_generated  $\sim$  tens;  $\geq 90\%$  belong to BinOp / Compare / 12 default classes; small killed-by-test\_a1.py ratio (PUT unit tests are output-shape/type sanity checks; most cosmic-ray mutants not killed), an empirical manifestation of the semantic-mutation criteria in the main manuscript: syntactic-tool mutants are not aligned with MR-violation detection. The 12-PUT generalisation (the main manuscript’s AST-overlap audit) supersedes this pilot but corroborates the same conclusion at scale.

#### B.5 Expected Root-Cause Risk Profile

**PUT-class  $\times$  LRCA-layer risk weights.** A numeric: C2 dominant ( $\star\star\star$  on L1 tolerance). B probabilistic: C4 dominant ( $\star\star\star$  on L3). C surrogate: C3 dominant ( $\star\star\star$  on L2 OOD;  $\star\star$  tolerance). D ML: C3 + C4 mixed ( $\star\star\star$  L2;  $\star\star$  L3).

**Operator-PUT root-cause hotspots (expected).** A:  $\text{mut}_{\text{inv}}/\text{mut}_{\text{conv}}/\text{mut}_{\text{cmp}}$  dominantly C2 (numerical tolerance),  $\text{mut}_{\text{mono}}/\text{mut}_{\text{dyn}}$  C1 (true semantic). B: dominantly C4 (statistical-assumption violations). C:  $\text{mut}_{\text{inv}}/\text{mut}_{\text{mono}}$  dominantly C2/C1 on c1/c2,  $\text{mut}_{\text{dyn}}/\text{mut}_{\text{cmp}}$  dominantly C3 on c1/c2; c3 (NN-Surr) row dominantly C5 (artefact). D: d1 row dominantly C5 (training-noise artefact); d2/d3 dominantly C1 / C3 mixed.

**Expected suspect\_share thresholds.** A: 0.10-0.20 (acceptance  $\leq 0.25$ ); B: 0.20-0.35 ( $\leq 0.40$ ); C: 0.20-0.30 ( $\leq 0.35$ ); D: 0.25-0.40 ( $\leq 0.45$ ). 60-cell average:  $\leq 0.20$  (= H4; acceptance  $\leq 0.25$ ).

## B.6 Engineering-significance mapping

$j = k$  aligned diagonal is the H2 threshold-test slice;  $j \neq k$  off-diagonal is control; 9 vacant cells (e, no MR exercised) are not formally adjudicated and are repurposed in the main manuscript’s R\_sem/R\_kill decoupling discussion as descriptive evidence for R\_sem / R\_kill decoupling. The  $v3 \rightarrow v4$  micro-change ( $\Delta\delta = -0.009$ , paired-role bootstrap 95% CI [-0.238, 0.207]) attributes to LLM-source diversity under a fixed prompt; this is the source-axis null contrast underpinning the main manuscript’s aligned-versus-cross results.

## B.7 Default-Operator Overlap between Syntactic Mutation Tools

Reviewer-requested manual operator-class cross-reference between cosmic-ray’s 13 default operators (cosmic-ray 8.4.6, cosmic\_ray.operators package) and mutmut’s 14 default operators (mutmut current main, src/mutmut/mutation/mutators.py). Both default-operator sets are first-order AST-local replacements; neither implements a cross-function-boundary, domain-knowledge-dependent, or algorithmic-class-changing mutation (the three necessary conditions in the semantic-mutation criteria of the main manuscript for semantic mutation).

Table 5. Mutmut vs cosmic-ray default-operator overlap.

| Operator class                              | cosmic-ray default                   | mutmut default                                              | Reaches the semantic-mutation criteria (a/b/c)? | Semantic-class reachability                           |
|---------------------------------------------|--------------------------------------|-------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------|
| Numeric literal $\pm 1$                     | number_replacer                      | operator_number                                             | (a) no, (b) no, (c) no                          | partial CE only (e.g. $\_RHO=28.0 \rightarrow 27.5$ ) |
| Binary arithmetic swap (+, -, *, /)         | binary_operator_replacement          | operator_swap_op (binary subset)                            | (a) no, (b) no, (c) no                          | partial OS only (incidental)                          |
| Comparison swap (<, <=, >, >=, ==, !=)      | comparison_operator_replacement      | operator_swap_op (compare subset)                           | (a) no, (b) no, (c) no                          | none                                                  |
| Boolean swap (and ++ or)                    | boolean_replacer                     | operator_swap_op (bool subset)                              | (a) no, (b) no, (c) no                          | none                                                  |
| Unary remove / swap                         | unary_operator_replacement           | operator_remove_unary_ops, operator_swap_op                 | (a) no, (b) no, (c) no                          | none                                                  |
| Keyword swap (True/False, etc.)             | keyword_replacer                     | operator_keywords, operator_name                            | (a) no, (b) no, (c) no                          | none                                                  |
| break ++ continue / return                  | break_continue                       | operator_keywords (subset)                                  | (a) no, (b) no, (c) no                          | none                                                  |
| Exception class swap                        | exception_replacer                   | (none)                                                      | (a) no, (b) no, (c) no                          | none                                                  |
| Decorator removal                           | remove_decorator                     | (none)                                                      | (a) no, (b) no, (c) no                          | none                                                  |
| Variable-binding insert / replace           | variable_inserter, variable_replacer | operator_arg_removal, operator_assignment                   | (a) no, (b) no, (c) no                          | none                                                  |
| no_op insertion                             | no_op                                | (none)                                                      | (a) no, (b) no, (c) no                          | none                                                  |
| Zero-iteration for loop                     | zero_iteration_for_loop              | (none)                                                      | (a) no, (b) no, (c) no                          | none                                                  |
| String content tweak (XX prefix, case flip) | (none)                               | operator_string                                             | (a) no, (b) no, (c) no                          | none                                                  |
| Lambda body $\rightarrow$ None/0            | (none)                               | operator_lambda                                             | (a) no, (b) no, (c) no                          | none                                                  |
| Dict kwarg name (XX prefix)                 | (none)                               | operator_dict_arguments                                     | (a) no, (b) no, (c) no                          | none                                                  |
| Sym./asym. string-method swap               | (none)                               | string-method symmetric swap; string-method asymmetric swap | (a) no, (b) no, (c) no                          | none                                                  |
| Augmented $\rightarrow$ simple assignment   | (none)                               | operator_augmented_assignment                               | (a) no, (b) no, (c) no                          | none                                                  |
| match case removal                          | (none)                               | operator_match                                              | (a) no, (b) no, (c) no                          | none                                                  |

**Aggregate.** Cosmic-ray  $\cap$  mutmut: ~9 of 13 cosmic-ray operators have a near-equivalent in mutmut (numeric, binary, compare, boolean, unary, keyword, break-continue, variable, plus partial overlap on string-prefix vs XX); 4 cosmic-ray operators are unique (exception, decorator, no\_op, zero-iteration-for); 6 mutmut operators are unique (string content, lambda, dict-kwarg, string-method swap, aug-assignment, match-case). The union is approximately 21 distinct first-order AST-local operator classes. None of the 21 can:

- (1) Cross a function-call or module-import boundary (e.g. replace `det(M)` with `sum(np.diag(M))` requires knowing the equivalence on diagonal matrices, outside any default-operator scope);
- (2) Depend on domain knowledge for legality (e.g. a hyperparameter swap that knows `noise_level=1e-4` is the load-bearing knob in a Gaussian Process Regression PUT requires PUT-class awareness);

- (3) Change the algorithmic class (e.g. swap polynomial  $\rightarrow$  spline basis rewrites the surrogate’s mathematical structure).

The mutmut / cosmic-ray null intersection on the main manuscript’s AST-overlap audit’s HP/SI/TF classes (zero overlap) therefore reflects a **structural property of first-order AST-local mutation** rather than a tool-specific limitation, supporting the preventive-defence framing in the main manuscript.

## C Experimental Procedure Details

### C.1 Cross-Source Generation Protocol

The per-cell identifiers used below (for example a2\_OS1, c1\_CE1, c3\_HP1) carry historical internal campaign suffixes CE/OS/HP/TF/SI, which name edit-mechanism families and are retained here only for reproducibility; they are not reader-facing semantic classes.

#### Two-stage ablation.

Table 6. Cross-source generation protocol.

| Version | Mutant pool                                                     | c-class primary MR              | Use                          |
|---------|-----------------------------------------------------------------|---------------------------------|------------------------------|
| v3      | Single-source (Claude Opus 4.6)                                 | P-ord. (pre-registered)         | H2 baseline                  |
| v4      | <b>Cross-source</b> (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat) | P-ord. (held at pre-registered) | Isolate LLM-source diversity |

**Protocol** (scripts/cross\_source\_campaign.py). (a) For each (PUT, operator), Claude / GPT / DeepSeek run  $K = 3$  trials with identical prompt (the cross-source protocol in the main manuscript), temperature 0.7; source\_tag propagated to filenames. (b) **V1-V4 mechanical validation** (src/p2/mutators/validation.py): V1 syntax (ast.parse), V2 executability, V3 non-triviality ( $|y_{\text{mutant}} - y_{\text{original}}| > 1e-6$ ), V4 signature consistency. (c) **DeepSeek model**: deepseek-chat (113 tokens/call, 1.8 s), not deepseek-v4-pro (340 tokens/call, 11 s), quality-equivalent on a2\_OS1 dry-run. (d) **Pool capacity**: 37 ops  $\times$  3 sources  $\times$  3 trials = 333 attempts; V1-V4 pass rate 89.5%; 298 confirmed mutants, of which 292 enter the v4 pool after final-stage rejection of duplicates and QA failures. Sources contribute nearly equally: Claude 101 / GPT 98 / DeepSeek 99 (Phase A engineering finding). (e) **Sampling** (scripts/build\_pools.py, POOL\_VERSION = v4): per-PUT 30 max, measured mean 24.3, range 10-30. c1 (GPR) has only 10 because c1\_HP1 / c1\_CE1 have V1-V4 near-zero pass (WhiteKernel noise\_level  $1e-4 \rightarrow 1e-1$  perturbation has minimal output impact, triggers V3 non-trivial failure, which supports the main manuscript’s R\_sem/R\_kill decoupling discussion).

**Protocol asymmetry**. v4 does not invoke reviewer LLM (cost / speed priority; a follow-up study will rerun dual-blind on the v4 grid, ~\$5-8 USD); v3 used the original Phase-1 dual-blind (Claude gen + GPT-5.4 review + DeepSeek arbitration). A fraction of the v3  $\rightarrow$  v4 source-axis shift may be slight v4 quality decline rather than LLM-source diversity.



## C.2 Differential prompt protocol

The  $v3 \rightarrow v4$  micro-change of -0.009 (paired-role bootstrap 95% CI [-0.238, 0.207]) may reflect “three LLMs under identical prompt” rather than “true upper bound of LLM diversity”. Separation requires *one tailored differential prompt per LLM* (skeleton: scripts/run-differential-prompt.py).

**Design (2×3 factorial, within-(PUT, operator)).** Factor A: prompt template, 3 levels: **V\_canonical** (control, identical to v4), **V\_persona** (per-LLM identity: Claude “numerical-analysis editor”; GPT “scientific-software refactoring”; DeepSeek “library-API substitution specialist”), **V\_cot** (Claude <thinking> tags; GPT “step by step”; DeepSeek stepped-reasoning). Factor B, LLM source (Claude / GPT-5.4 / DeepSeek chat).  $K = 3$  per cell; total  $37 \times 3 \times 3 \times 3 = 999$  trials.

**Exit criteria.** If  $\max(\text{delta}) - \min(\text{delta}) < 0.05$ : the  $v3 \rightarrow v4 - 0.009$  source-axis null is robust. If  $\geq 0.05$  with prompt variant pushing delta past 0.474: H2 revised from “not met” to “prompt-conditional met”. If  $\geq 0.05$  but all delta  $< 0.474$ : prompt sensitivity exists but H2 unchanged.

**Resources.** API ~\$18-30 USD; wall 3-4 h at concurrency 4; V1-V4 only, no reviewer LLM.

## C.3 Manual Pilot and Initial Prescreen

The original cross-source protocol used 60% multi-LLM consensus (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat unanimous) plus 40% manual injection by scientific-computing researchers. Each (mut\_j, S\_i) pair carried one prompt template containing PUT source, semantic intent of mut\_j, prohibition against revealing specific MRs (avoids prompt leakage), and output requirements (syntactically correct, executable, single-point modification, diff < 10 lines). Reproducibility parameters: temperature 0.3 in manual-pilot stage (v4 cross-source uses 0.7 per §C.1); fixed seed; 5 candidates per prompt with 2-3 retained.

**Double-blind review (Scheme C).** Generator LLM-G (Claude Opus) and reviewer LLM-R (GPT-5.4) are cross-source with strict role separation. LLM-R sees only PUT original + mutant code, unaware of mut\_j category / MR content / generator identity, outputting a triple (syntactic, executable, semantic-fault-injected). Double-confirmed mutants enter pool; inconsistencies enter manual arbitration ( $\leq 10\%$ ). From the double-confirmed pool, 20% is sampled for manual review; manual-vs-LLM-R inconsistency  $> 10\%$  triggers full manual downgrade. Same-source bias mitigation stratifies LLM-R by PUT class. Prompt-injection mitigation disables RAG / web search on the API path.

**L0 pool prescreen.** Each mutant passes static syntax (linters / type checkers), a unit self-test (simple inputs produce finite output), and double-blind sign-off. Target: 30-50 candidates per cell  $\rightarrow$  10-15 retained after prescreen.

## C.4 Three-Layer Root-Cause Diagnosis

Decision tree, multi-label priority ( $C5 > C4 > C3 > C2 > C1$ ), and output (C1\_share, suspect\_share) are stated in §A.2; the full pseudocode is reproduced there. LRCA does **not** modify SMS; killed set is not filtered. The main manuscript’s LRCA-mass results use the best calibrated combination (ood\_band = 0.02, tolerance\_multiplier = 3.0, repeats = 20); default-threshold results retained in lrca\_60cell\_v3.json as control. Interface with the preventive-defence risk profile in the main manuscript: L0 prescan  $\leftrightarrow$  §C.2 dual-LLM double-blind; L1 tolerance  $\leftrightarrow$  N = 20 repetition subprocess; L2 OOD  $\leftrightarrow$  input-distribution definition; L3 statistical baseline  $\leftrightarrow$  pre-check protocol.

## C.5 Pilot calibration

A 2026 Q3 operator-level pilot ran 12 PUTs  $\times$  37 operators (36 primary PUT-class operator pairs plus one control-flow structural-injection specialisation; the conceptual semantic family count remains five; 12 is\_key=True at K=20, 25 at K=10; 470 trials total). Pilot precursor quantities: R\_sem (V1-V4  $\wedge$  operator\_match pass rate), D\_impl (median pairwise AST + literal + identifier Jaccard distance among confirmed mutants), R\_kill (proportion of confirmed mutants killed by AVP under that PUT's primary MR). Aggregate (N=37): R\_sem median 0.50 mean 0.468 (0/37 with R\_sem=0); D\_impl median 0.42 mean 0.392 (1/37 with D\_impl  $\approx$  0); R\_kill median 0.00 mean 0.189. All 37 operators produced  $\geq 1$  V1-V4 passing mutant, a necessary precondition for H1; the pre-registered H1 threshold itself ( $\geq 5$  non-equivalent mutants on  $\geq 9/12$  PUTs for  $\geq 4$  of 5 operator families) is nonetheless not met, as reported in the main text.

**Key R\_sem / R\_kill decoupling pattern.** R\_sem  $\geq 0.9 \wedge$  R\_kill = 0 combinations all on HP / TF / OS operators in c / d classes (e.g., c1\_CE1 noise 1e-4 $\rightarrow$ 1e-1 with R\_sem 1.00, R\_kill 0.00; c3\_HP1 relu $\rightarrow$ tanh, c3\_TF1 max\_iter 1000 $\rightarrow$ 5, d1\_HP1 MLP  $\alpha$ , d3\_HP1 LR C). R\_sem-high  $\wedge$  R\_kill = 1 combinations all on CE / OS operators in a / b classes (a2\_CE1 LU det, a2\_OS1 prod $\rightarrow$ sum, b1\_OS1  $\alpha/\beta$  swap, d1\_TF1 label flip). This is empirical evidence for R\_sem / R\_kill decoupling at cell level.

## D Statistical Analysis Details

### D.1 Semantic-Mutation Score Variants and Construction Lemmas

**SMS\_unfiltered (reviewer comparison metric):**

$$\text{SMS}_{i,k,j}^{\text{unfiltered}} := \frac{|\text{killed}_{i,k,j}|}{|\text{mut}_j(S_i)| - |\text{equiv}_{i,k,j}|}$$

(Same form as the primary metric; here killed does not distinguish C1 from C2-C5.)

The appendix of the reproducibility package provides cell-by-cell difference between SMS\_unfiltered and SMS; if relative difference  $< 5\%$ , this confirms LRCA does not affect robustness of primary conclusions.

**Primary-table construction.**

Table 7. SMS variants and construction lemmas.

| RQ          | Primary statistics                                                     | Reporting format                              |
|-------------|------------------------------------------------------------------------|-----------------------------------------------|
| RQ3         | inst_rate, equiv_rate, C1_share, survive_rate                          | 60-cell heatmap + 4-class marginals           |
| RQ4         | aligned-SMS vs cross-SMS; sparse $\circ$ vs dense $\bullet$ equiv_rate | Cliff's delta + odds ratio + 95% bootstrap CI |
| RQ5         | $\Delta\text{SMS}_c$ ( $c \in \{A,B,C,D\}$ ); CV( $\Delta\text{SMS}$ ) | sign test (df=3) + descriptive forest plot    |
| RQ5 (desc.) | Spearman $\rho$ + Kendall $\tau$ for SMS vs PC                         | scatter + dual-metric ranking comparison      |

**Multiple comparisons.** No cell-level family of p-values is claimed across the 60 cells (main text, statistical pipeline); the per-class Friedman tests carry a Bonferroni  $\times 4$  correction.  $N = 20$  bootstrap intervals: 1,000-iteration 95% CI ( $B = 10,000$  for the headline H2 delta CI).

## D.2 Effect-Size Sensitivity and Transformation Invariance

**H4 cutoff sensitivity.** Dense grid scan (cutoff  $\in \{0.05, 0.10, \dots, 0.50\}$ , step 0.05) on v4 data (scripts/h5\_sensitivity.py, data/results/h5\_sensitivity\_v4.json):

Table 8. Effect-size sensitivity and logit-transformation invariance.

| cutoff              | h5_cells_pass  | h5_pass_ratio |
|---------------------|----------------|---------------|
| 0.05                | 12 / 60        | 20.0%         |
| 0.10                | 12 / 60        | 20.0%         |
| 0.15                | 12 / 60        | 20.0%         |
| <b>0.20 (paper)</b> | <b>12 / 60</b> | <b>20.0%</b>  |
| 0.25                | 12 / 60        | 20.0%         |
| 0.30                | 12 / 60        | 20.0%         |
| 0.35                | 12 / 60        | 20.0%         |
| 0.40                | 12 / 60        | 20.0%         |
| 0.45                | 13 / 60        | 21.7%         |
| 0.50                | 13 / 60        | 21.7%         |

**Conclusion.** H4 verdict (not met) is intrinsic data property, independent of cutoff choice. v4 suspect\_share distribution is severely bimodal: median 1.0, mean 0.79; 48 cross cells at suspect\_share  $\approx 1$ , 12 aligned cells in low end; the  $[0.20, 0.80]$  interior is nearly empty. Specific value 0.20 is **not load-bearing**.

**Logit-transformation invariance.** Cliff's delta is rank-based (function of  $U/(n_1 \cdot n_2)$ ), mathematically invariant under any strictly monotone transformation (Romano 2006). Logit is strictly monotone on  $(0, 1)$ , so  $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$  is a construction result. **Not additional robustness evidence;** H2 robustness threats come from the main manuscript's 60-cell distribution results zero-mass dominance, not metric-scale choice.

**Vacant-cell sensitivity.** Excluding the 9 vacant cells (cells with no exercised MR) leaves the aligned-versus-cross verdict unchanged. In v4 the analysis becomes  $n = 11$  aligned and  $n = 40$  cross with  $\delta = 0.323$  (95% CI  $[-0.011, 0.648]$ ); the matching v3 slice gives  $\delta = 0.270$  (95% CI  $[-0.043, 0.595]$ ). The exclusion widens the interval because the design is smaller, but it does not move H2 across the pre-registered Romano medium threshold.

**Primary-MR withdrawal permutation-null.** The auxiliary c-class check pools 15 SMS values, shuffles labels over the  $3 \times 5$  (PUT, MR) grid, and applies the max-over-five withdrawal rule with 10,000 permutations (seed 42). The observed c-class mean is 0.314 versus null mean 0.349, one-sided  $p = 0.989$  (data/results/c\_class\_permutation\_v4.json). This supports treating the primary-MR convention as a protocol choice rather than as evidence of a hidden c-class inflation effect.

### D.3 Power Analysis under Observed and Stipulated Effects

**Plug-in bootstrap.** With-replacement sample from observed aligned (n=12) and cross (n=48) v4 SMS pools;  $N_{\text{sim}} = 5,000$ , seed = 42 (scripts/compute\_rq2\_power.py, rq2\_power\_v4.json):

Table 9. Power analysis under observed and stipulated effects.

| Threshold               | Interpretation    | Power        |
|-------------------------|-------------------|--------------|
| delta > 0.000           | Any-effect        | <b>0.997</b> |
| delta > 0.147           | Small             | 0.966        |
| delta > 0.330           | Medium            | 0.759        |
| <b>delta &gt; 0.474</b> | <b>Large (H2)</b> | <b>0.423</b> |

Sample-size sweep ( $n_{\text{aligned}} \in \{6, 12, \dots, 60\}$ ,  $n_{\text{cross}} = 4 \times n_{\text{aligned}}$ ): power for delta > 0 reaches 0.974 at n=6, 0.996 at 12, plateaus thereafter. RQ4 primary analysis has very low sample-size requirement; the bottleneck is the effect-size boundary itself, not sampling noise. Even at  $\delta_{\text{truth}} \approx 0.474$ , sample has only ~42% chance of detecting under plug-in; but this does **not** mean “insufficient power is the cause of H2 not being met”, observed  $\delta_{v3} = 0.323 < 0.474$ , observed  $\delta_{v4} = 0.314 < 0.474$ ; increasing sample only narrows CI.

**Stipulated-alternative.** Implemented in mixture-weight (scripts/compute\_rq2\_power\_stipulated.py,  $N_{\text{sim}} = 2,000$ ): SMS has heavy ties at zero (45/60 = 0), raw shifting is discontinuous (any  $\varepsilon > 0$  jumps delta from 0.314 to 0.74). Mixture: aligned' = (probability w) sample from (observed\_aligned + 0.001) + (probability 1-w) sample from observed\_aligned, calibrate w so  $E[\text{delta}] \approx 0.474$ . Calibrated w = 0.094, realised  $E[\text{delta}] = 0.4746$ .

Table 10. Power analysis under observed and stipulated effects.

| Stipulated truth | Criterion                                  | Power        |
|------------------|--------------------------------------------|--------------|
| 0.474            | $\hat{\delta} \geq 0.474$ (point-estimate) | <b>0.491</b> |
| 0.474            | 95% CI lower > 0 (any-effect)              | 0.868        |

Even at  $\delta_{\text{truth}} = 0.474$ , the (12, 48) design returns  $\hat{\delta} \geq 0.474$  in only ~49% of replications. This *supports* the main manuscript’s aligned-versus-cross framing that “not met” is point-estimate, not effect-size.

### D.4 Per-Class Friedman Tests

Per-class Friedman (3 SUTs  $\times$  5 MRs) with Bonferroni  $\times$  4: a  $\chi^2=4.00$  raw 0.406 adj 1.000, W=0.333; b  $\chi^2=10.78$  raw **0.029** adj **0.116**, W=**0.898**; c  $\chi^2=4.00$  raw 0.406 adj 1.000, W=0.333; d  $\chi^2=5.00$  raw 0.287 adj 1.000, W=0.417. After correction, no per-class result remains significant at family-wise alpha = 0.05. Even b-class W = 0.898 (Cohen 1988 large concordance) is insufficient at N = 3 per class. **H3 verdict rests on the main manuscript’s cross-class consistency sign test**; per-class Friedman is sensitivity / descriptive only. Friedman main effect on full 60 cells ( $\chi^2 = 15.30$ , p = 0.0041) confirms MR rank differences but is logically independent from H3 cross-class consistency.

**Mixed-effects unavailability.** Primary sms ~ C(class) + C(operator) + C(class):C(operator) + (1 | put) returned Singular (insufficient column rank for class  $\times$  operator interaction; N=60 too small for 11-d fixed-effects

+ 12 PUT random intercepts). Fallback  $\text{sms} \sim C(\text{class}) + C(\text{operator}) + (1 \mid \text{put})$  had PUT random-intercept variance hit boundary 0 (degenerates to OLS). Fallback class p-values (descriptive only): b vs a 0.275 / c vs a 0.892 / d vs a 0.991. RQ5 conclusion shifts to four-piece presentation (class means + sign test + Friedman + forest plot).

## D.5 Pattern-coverage operationalisation

Per SUT, ( $\text{MR\_k}, \text{R\_outcome} \in \{\text{True}, \text{False}\}$ ) binary-tuple coverage:  $5 \text{ MRs} \times 2 \text{ outcomes} = 10 \text{ cells}$ ,  $\text{PC} = \# \text{triggered} / 10$ . Simplest baseline (per-SUT granularity, not distinguishing mutants). PC range over 12 SUTs: [0.500, 1.000], mean 0.733 (paper\_numbers\_v4.json). Pairing per PUT: Spearman  $\rho = 0.163$  ( $p = 0.613$ ); Kendall  $\tau = 0.136$  ( $p = 0.568$ ).

**Power caveat.** At  $n = 12$ , Spearman 95% CI  $\approx [-0.5, +0.6]$  (Fisher-z approximation); cannot distinguish “zero correlation” from “moderate positive” or “moderate negative”.  $p = 0.61 / 0.57$  means “no correlation detected at  $n = 12$ ”, not “correlation does not exist”.

**Within-class descriptive.** b2 PC = 1.0 with mean SMS = 0.067; b1 PC = 0.7 with mean SMS = 0.20. c3 PC = 1.0 with mean SMS = 0.14; c1 / c2 PC = 0.7-0.8 with mean SMS = 0.0. Within the same class, higher-PC PUT has lower SMS, contradicts naive “more PC kills more mutants” assumption. This pattern is consistent with orthogonality, but  $n = 12$  cannot confirm it. A confirmatory study would need  $n \geq 30$  and a PC definition that incorporates the mutant dimension.

## E Stakeholder Deployment Considerations

### E.1 Air-Gap Deployment Constraints

The stakeholder-analysis workflow in the main manuscript depends on external LLM API calls (Claude / GPT / DeepSeek), incompatible with most regulated air-gapped V&V workflows:

Table 11. Air-gap deployment constraints.

| Domain     | Standard  | Air-gap requirement                                                               |
|------------|-----------|-----------------------------------------------------------------------------------|
| Nuclear    | IEC 60880 | Air-gapped build for safety-critical software                                     |
| Aerospace  | DO-178C   | Tool-qualified offline build for DAL-A/B; LLM API outside qualified-tool envelope |
| Medical    | IEC 62304 | Class C software lifecycle requires offline reproducible build                    |
| Automotive | ISO 26262 | ASIL-D requires offline tool chain; cloud LLM non-compliant                       |

**Mitigation paths.** (a) Self-hosted open-weight LLMs (Llama / DeepSeek local inference), capability gap vs API-served frontier models; quantitative impact on SMS untested. (b) Offline-cached pre-generated mutant pools with commit-hash + raw-response signature locking, adopters reproduce a published pool offline (this paper’s package supports this), but generating new pools requires external LLM access. SMS for air-gapped industrial deployment is a future research direction.

## E.2 Quarterly batch audit workflow

The original per-PR GitHub Actions YAML was removed because its PR-CI threshold 0.10 contradicted the Appendix E.3 audit threshold 0.20 and propagated stakeholder-facing CI assumptions into adopters' pipelines.

### Replacement: quarterly batch audit:

Table 12. Quarterly batch audit workflow.

| Step | Description                                                                                      | Cost                     |
|------|--------------------------------------------------------------------------------------------------|--------------------------|
| 1    | Offline mutant-pool generation (per PUT $24 \times 30 \times 12$ PUTs $\approx 300$ )            | LLM API \$5-15; ~30 min  |
| 2    | <code>sms_campaign.py --track 2</code>                                                           | 4-core laptop ~10-20 min |
| 3    | <code>run_lrca.py</code>                                                                         | < 1 min                  |
| 4    | MR-design team review of MRs with aligned-cell SMS significantly below historical median (0.275) | 1-2 h meeting            |

Quarterly: API ~\$10 + automation 30 min + manual 1-2 h  $\approx$  0.5 person-day. Affordable. Per-PR gating not recommended (LLM latency 5-30 s + cost + non-determinism + air-gap incompatibility); adopters needing per-PR should use coverage / unit-test gates. SMS does not replace domain-expert MR physical-reasonableness judgement nor product-requirements review.

## E.3 Verification-and-Validation Documentation

This appendix subsection makes **no normative claim** toward NRC, FDA, or ISO 26262 review teams. Within single-output kernels scope, no traceable mapping exists to current IEC 60880 / ISO 26262 / DO-178C / ASME V&V 20-2009 normative bodies.

V&V documentation for scientific computing software (per ASME V&V 20-2009 §3 and similar guides) requires quantifiable test-adequacy evidence; current documentation often relies on code coverage + MR lists + SME signatures. SMS may appear as **research-grade supplementary evidence** alongside coverage and MR lists: (a) aligned-cell SMS per critical PUT; (b) LRCA three-layer diagnosis (C1 readout / C2-C5 attribution); (c) 60-cell matrix visualisation. Reviewers can independently run `REPRODUCIBILITY.md`.

Original threshold recommendations (aligned-cell SMS  $\geq 0.20$  /  $0.30 + C1\_share \leq 0.20$ ) were removed: no normative backing in IEC / ISO / ASME; misreads as enforce-ready. **SMS reported as descriptive supplementary evidence**, with adequacy judgements left to V&V reviewers case-by-case.

Conceptually complementary to ASME V&V 20-2009 §3 code verification (numerical-solver correctness) vs SMS (MR-set fault-detection adequacy). Substantial scale gap to multi-module CFD codes; incorporation into V&V standards body would require large-scale empirics + multi-year dialogue with ASME V&V 20 / IEEE 1012 / IEC 60880 committees. **No advocacy that SMS enter any normative certification system in 2027.** SMS does not replace SME signatures, FMECA, PIRT, system-level V&V, or ASME V&V 20 §3 numerical-solver verification, it is one link in the evidence chain, not a single-point qualification.

## F Threats to Validity: Detailed Mitigation

### F.1 Internal threats

**Hoeffding-style false-equivalence bound (referenced by main-text Limitation 1).** For a mutant whose true per-sample disagreement probability against the original program under  $D_S$  is at least  $p_0$  (at the E2 tolerance), the probability that all  $K_{eq}$  independent E2 samples agree, producing a false equivalence verdict, is at most  $(1 - p_0)^{K_{eq}} \leq \exp(-K_{eq} p_0)$ . At  $K_{eq} = 1000$ :  $p_0 = 0.005$  gives  $\leq e^{-5} \approx 0.7\%$ ;  $p_0 = 0.01$  gives  $\leq e^{-10} \approx 4.5 \times 10^{-5}$ . The bound conditions on  $p_0$  and says nothing about mutants whose disagreement is concentrated below the E2 tolerance; those are governed by the E1 arm and remain the residual risk the K\_eq sweep (deferred) would quantify.

Table 13. Internal threats.

| Threat area               | Threat                           | Mitigation                                                                                                                                                                    | Residual                                                     |
|---------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| LLM generation            | LLM generation reproducibility   | §C.2 reproducibility parameters in package; dual-LLM cross-source + 20% manual; reproducibility rests on archived raw responses and frozen mutant pools (no user-facing seed) | LLM training data may be updated post-submission             |
| Equivalence approximation | Probabilistic equiv (K_eq=1000)  | the main manuscript’s equivalence-judgement section declares probabilistic approximation; Hoeffding-style false-equiv bound                                                   | K_eq sweep $\in \{500, 1000, 2000\}$ deferred to future work |
| Shared infrastructure     | Shared-infrastructure dependency | AVP version pinned to the upstream commit hash; package embeds AVP source; the arXiv infrastructure technical report is disclosed in the main text                            | None within scope                                            |
| Root-cause diagnosis      | LRCA multi-label boundary        | §A.2 priority C5 > C4 > C3 > C2 > C1; multi-label co-occurrence table in package; root_cause is “likely root cause” not definitive                                            | C2-C5 may multi-trigger on B/D classes                       |

|                           |                                             |                                                                                                                                              |                                                                          |
|---------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Pool size                 | Mutant-pool size                            | Pool expanded to mean 17.4: delta moved 0.321 → 0.323, CI narrowed; <b>effect size is intrinsic ceiling, not pool dilution</b>               | Some PUTs < 30 (cache-bound)                                             |
| LLM non-determinism       | LLM non-determinism                         | Multi-turn de-dup; K=10/20 repetitions; raw prompt+response committed for direct reuse                                                       | Claude Opus subscription lacks seed control                              |
| <b>Protocol asymmetry</b> | <b>Protocol-implementation gap v3 vs v4</b> | v4 used 3 providers (Claude 101 / GPT 98 / DeepSeek 99) without dual-blind; v4 C1_share 0.209 > v3 0.164 weakly opposes “v4 quality decline” | Complete separation requires a follow-up dual-blind rerun on the v4 grid |

The protocol-asymmetry threat (dual-blind vs V1-V4 only) applies to the v3 → v4 source-axis contrast ( $\Delta\delta = -0.009$ , paired-role bootstrap 95% CI [-0.238, 0.207]).

## F.2 External / construct / conclusion threats

Table 14. External / construct / conclusion threats.

| Threat area        | Threat                        | Class      | Mitigation                                                                                                                       |
|--------------------|-------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------|
| Representativeness | 12-PUT representativeness     | External   | follows the shared site selection; open class principle (cls extensible); scaling to MD/QC/CFD                                   |
| Statistical power  | Cross-class statistical power | Conclusion | H3 framed exploratory; mixed-effects unavailable (Singular at N=60); shifted to class means + sign test + Friedman + forest plot |
| LLM homogeneity    | LLM homogeneity bias          | Construct  | §C.2 three-LLM rotation + PUT-class subdivision + 20% manual sampling; third-party LLM family validation deferred                |



|                       |                                 |          |                                                                                                                                                                                                                                          |
|-----------------------|---------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LLM-source shift      | LLM-source distributional shift | External | Hit distribution DeepSeek 11/15, Claude 4/15, GPT 0/15; the main manuscript’s AST-overlap audit argument is categorical AST-locality, not hit frequency; the zero-overlap edit-mechanism families remain unaffected across all 3 sources |
| Higher-order mutation | HOM equivalence                 | External | Tool-unreachability claim confined to first-order syntactic tools; HOM empirical testing listed as future work                                                                                                                           |

Conclusion threats: multiple comparisons handled per the main-text policy (per-class Bonferroni  $\times 4$ ; no cell-level family claimed; §D.1);  $N=20$  stability handled by 1,000-iteration bootstrap CI (§D.3). Construct threat, SMS measures epistemological semantic detection, not production engineering value (out of scope here).

## G Degeneration to Classical Mutation Score: Full Proof

### G.1 Notation cross-reference

PUT  $S_i$ , mutation operator family  $\text{mut}$  (syntactic or semantic), mutant set  $\text{mut}(S_i)$ , MR-label set, equivalence tolerance  $\epsilon_{\text{eq}}$ , equivalence sample size  $K_{\text{eq}}$ , AVP tolerance  $\epsilon_{\text{AVP}}$ . Three-state decomposition:  $\text{mut}(S) = \text{killed} \cup \text{equiv} \cup \text{survive}$  (disjoint). SMS formula:

$$\text{SMS}_{i,k,j} = \frac{|\text{killed}_{i,k,j}|}{|\text{mut}_j(S_i)| - |\text{equiv}_{i,k,j}|}$$

### G.2 Degenerate-limit definition

The degenerate limit  $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$  consists of three **joint conditions** (paired axes), each controlling one layer of the SMS formula. L1-L6 are not 6 independent axes; the pairing reflects dependency structure across the SMS formula layers.

Table 15. Degenerate-limit definition.

| Joint condition                 | Component axes                                                                                                                                         | Pairing rationale                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $L_{\text{equiv}}$ (Lemma G.1)  | L1: $\varepsilon_{\text{eq}} \rightarrow 0$ ; L2: $K_{\text{eq}} \rightarrow \infty$                                                                   | L1 alone: equiv stays probabilistic ( $K_{\text{eq}}$ cannot cover $D_S$ ). L2 alone: bitwise equality diluted by $\varepsilon_{\text{eq}}$ . Both simultaneously degenerate equiv to classical behavioural equivalence outside a $D_S$ -measure-zero set.                                                                                                           |
| $L_{\text{killed}}$ (Lemma G.2) | L3: $\varepsilon_{\text{AVP}}^k \rightarrow 0$ ; L4: MR-label set = $\{\text{MR}_{\text{eq}}\}$ , $R(y, y') \equiv y = y'$                             | L3 alone: $\varepsilon_{\text{AVP}} \rightarrow 0$ still allows non-trivial MR (monotonicity, convergence). L4 alone: equality still carries $\varepsilon_{\text{AVP}}$ tolerance. Both simultaneously degenerate killed to classical difference detection.                                                                                                          |
| $L_{\text{mut}}$ (Lemma G.3)    | L5: $\text{mut}_j \rightarrow \text{rule-based syntactic}$ (Mothra AOR/ROR/SDL/CRP); L6: $\text{cls}(I) \subseteq \{\text{imperative deterministic}\}$ | L5 alone: syntactic ops on probabilistic/ML may still trigger domain-semantic subsets. L6 alone: imperative programs are still mutable by non-AST-local edit-mechanism operators (API-substitution, hyperparameter, numerical-method-order, and structural-injection families). Both simultaneously degenerate $\text{mut}(S)$ to Jia & Harman syntactic mutant set. |

### G.3 Lemmas for Three-State Decomposition

**Lemma G.1 (equiv degeneration; measure-zero qualification).** Under  $L_{\text{equiv}}$  ( $L1 \wedge L2$ ), semantic-class equivalence ( $E1 \wedge E2$ ) degenerates to classical behavioural equivalence **almost everywhere** w.r.t. measure  $D_S$ .

*Proof.* E1 (type consistency) holds trivially under  $\varepsilon_{\text{eq}} \rightarrow 0$  (L6 makes imperative output spaces scalar/vector with static types). E2 ( $|S_i(x) - s'(x)| < \varepsilon_{\text{eq}}$  for  $K_{\text{eq}}$  samples) under  $L1$  ( $\varepsilon_{\text{eq}} \rightarrow 0$ )  $\wedge$   $L2$  ( $K_{\text{eq}} \rightarrow \infty$  with measure-equivalent sampling) is almost-everywhere equivalent to  $\forall x \in D_S \setminus N : S_i(x) = s'(x)$ , where  $N$  is a  $D_S$ -measure-zero set (continuous  $D_S$ : floating-point pathological points / NaN propagation; discrete  $D_S$ :  $N = \emptyset$ ). This matches Jia and Harman [1] §3 classical equivalent-mutant definition under measure-zero equivalence classes.

**Lemma G.2 (killed degeneration).** Under  $L3 \wedge L4$ , killed degenerates to classical difference detection.

*Proof.* L4 restricts the MR-label set to  $\{\text{MR}_{\text{eq}}\}$  with  $R(y, y') \equiv y = y'$ . For  $\text{mr} = (r, R)$  and mutant  $s'$ , the  $\text{MR}_{\text{eq}}$  violation condition  $\exists x : S_i(x) \neq s'(r(x))$  under L3 ( $\varepsilon_{\text{AVP}} \rightarrow 0$ ) becomes exact inequality. With  $r = \text{id}$ , violation is  $S_i(x) \neq s'(x)$ , classical difference detection. With  $r \neq \text{id}$ ,  $\text{MR}_{\text{eq}}$  still requires  $S_i(x) = s'(r(x))$  as reference oracle from original program; no new state classifications introduced.

**Lemma G.3 (mut degeneration).** Under  $L5 \wedge L6$ ,  $\text{mut}_j(S_i)$  degenerates to the syntactic mutant set of Jia and Harman [1].

*Proof.*  $L5$  switches  $\text{mut}_j$  to rule-based syntactic operators (AOR, ROR, SDL, CRP, UOI, standard Mothra/Proteum sets);  $L6$  restricts PUTs to imperative deterministic programs, excluding triggering conditions for semantic operators on probabilistic/ML programs. Under this configuration,  $\text{mut}_j(S_i)$  is the literature-defined syntactic mutant set, independent of domain semantics.

#### G.4 Degeneration Theorem: Detailed Proof

**Theorem G (SMS  $\rightarrow$  MS degeneration; stated in the main text as the degeneration theorem).** In the degenerate limit  $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$ , almost everywhere w.r.t.  $D_S$ ,

$$\text{SMS}_{i,k,j} \xrightarrow{L} \text{MS}_{i,j} := \frac{|\text{killed}_{i,j}^{\text{classic}}|}{|\text{mut}_j^{\text{syntax}}(S_i)| - |\text{equiv}_{i,j}^{\text{classic}}|}$$

*Proof.* Combine Lemmas 9.1-9.3:

- Numerator: under  $L3 \wedge L4$ ,  $\text{killed}_{i,k,j} \rightarrow \text{killed}_{i,j}^{\text{classic}}$  (Lemma G.2);  $L4$  makes  $\text{MR}_{i,k}$  trivial over  $k$  (only  $\text{MR}_{\text{eq}}$  remains), so subscript  $k$  degenerates.
- Denominator  $|\text{mut}_j(S_i)|$ : under  $L5 \wedge L6 \rightarrow |\text{mut}_j^{\text{syntax}}(S_i)|$  (Lemma G.3).
- Denominator  $|\text{equiv}_{i,k,j}|$ : under  $L1 \wedge L2 \rightarrow |\text{equiv}_{i,j}^{\text{classic}}|$  (Lemma G.1; almost-everywhere w.r.t.  $D_S$ ).

Substituting into the SMS formula gives  $\text{MS}_{i,j}$ .

**Corollary G (LRCA trivialisation).** Under  $L = L1 \wedge L2 \wedge L3$ ,  $C = \{C1, \dots, C5\}$  degenerates to  $\{C1\}$ .

*Sketch.* Each of  $C2$ - $C5$ 's triggering precondition depends on at least one  $L_j$  being violated. When  $L1 \wedge L2 \wedge L3$  hold simultaneously, every non-trivial-space dimension (MR non-triviality, AVP tolerance non-zero, non-empty equiv set, MR-design DOF, class-mapping openness) closes;  $C2$ - $C5$  triggering set becomes empty (read off from §A.2 decision tree). The per- $C_k$  to per- $L_j$  minimum-sufficient mapping depends on the main manuscript's LRCA-diagnosis engineering thresholds; we do not claim one-to-one correspondence at formal level, readers can trace through §A.2 + §C.4. Under  $L$ ,  $\text{suspect\_share} \rightarrow 0$ , LRCA reports only  $C1$ , SMS degenerates to single-layer metric consistent with Jia and Harman [1] MS engineering structure.

**Empirical consistency.** Theorem G and the LRCA-trivialisation corollary jointly show that the SMS-based empirical conclusions (Cliff's delta in the aligned-versus-cross results, Friedman  $\chi^2$  in the cross-class consistency results, and Spearman rho in the SMS-vs-Pattern-Coverage analysis) are structurally consistent with existing Jia and Harman [1] literature in classical syntactic-mutation scenarios, no metric-level semantic fragmentation.

#### H Adjoint Extension Study

This appendix gives the full specification of the confirmatory adjoint extension arm summarised in the main manuscript. The arm exercises the adjoint invariant  $\psi_6$  on three third-party linear-operator libraries, disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor:

- **scipy.sparse.linalg LinearOperator** (linear algebra; BSD-3). Program under test: the scaled-operator adjoint  $M^H$  for  $M = \alpha A$  with complex scale  $\alpha$ . Real anchor: scipy issue #8900 / PR #8962 (the adjoint dropped the conjugate on  $\alpha$ , returning  $\alpha A^H$  instead of  $\bar{\alpha} A^H$ ).

- **pylops.signalprocessing.FFT** with `real=True` (signal processing; BSD-3). Program under test: the real-FFT adjoint `_rmatvec`; the adjoint identity is the library’s own *dot-test*. Real anchor: pylops issue #268 / PR #292 (commit bd3a919), a wrong scaling of the middle/Nyquist frequency bin that made the adjoint inconsistent with the forward.
- **jax.linear\_transpose / vjp** (automatic differentiation; Apache-2.0). Program under test: the autodiff-derived transpose  $M^T$  of a linear operator containing an inverse real FFT,  $M(x) = \text{irfft}(\text{rfft}(x) \cdot w)$ . Real anchor: jax issue #6223 / PR #6232 (commit 112ae41), an `irfft` transpose rule that reversed order with an off-by-one index, so `vjp` silently produced a non-adjoint transpose.

For each subject the MR families are `adj.self` (self-adjointness) and `adj.dual` (scaled/dual-operator transpose). The mutant pool spans adjoint-breaking fault mechanisms (drop-conjugate, sign flip, scale and phase errors, transpose-without-conjugation, Nyquist mis-scaling), with the real fixed defect included among them, plus semantically equivalent mutants (scalar-multiplication reorder, conjugation of a real adjoint output) that a strong MR must *not* kill. Because the pre-fix releases are uninstallable on the host platform (no `arm64` wheel), each real defect is reproduced in extracted-diff form on the current release, mirroring the boundary study;  $\epsilon_{\text{tol}}$  is an explicit factor throughout.

Table 16. Adjoint extension study, forward kill.

| Substrate            | Operator class (real anchor)      | Active killed / total | Equiv. survived | Forward SMS |
|----------------------|-----------------------------------|-----------------------|-----------------|-------------|
| scipy LinearOperator | linear algebra (#8900)            | 5 / 5                 | 2 / 2           | <b>1.00</b> |
| pylops FFT real      | signal processing (#292)          | 4 / 4                 | 2 / 2           | <b>1.00</b> |
| jax vjp / transpose  | automatic differentiation (#6223) | 4 / 4                 | 2 / 2           | <b>1.00</b> |

Across all three substrates the correct program survives, every non-equivalent (active) mutant is killed, including the three real fixed defects, and no semantically equivalent mutant is killed, so the forward SMS is 1.00 with zero false positives on the equivalent set. The `scipy` #8900 defect is shared with the strong-boundary study in the main manuscript (`adj.self` case) and is not counted as independent corroboration twice; the `pylops` #292 and `jax` #6223 defects are unique to this arm. The pools are hand-constructed and low-cardinality, so the arm demonstrates the duality forward rather than measuring a high-adequacy MR set on a discovered fault population (main-paper adjoint-arm reproduction-fidelity threat).

## I Real-Defect Evidence Summary

This appendix records only the result-level facts used by the main manuscript’s real-defect evidence slice. It deliberately excludes candidate mining, rejected cases, repository tooling, and benchmark-release narrative. The purpose is auditability for the semantic-mutation argument: kill-rate, semantic-stratum alignment, and real-defect detection are related but distinct constructs.

Table 17. Result-level real-defect evidence summary.

| Audit item               | Evidence content used in the manuscript                                                                                                                                                                                                                                | Main-text role                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Verified stratum counts  | 34 reproduced, MR-detectable defects (33 <code>verified_full</code> and one disclosed candidate case; see the count-caliber note) across five meta-patterns: $m_{\text{mono}}$ 11, $m_{\text{conv}}$ 8, $m_{\text{inv}}$ 6, $m_{\text{adj}}$ 5, and $m_{\text{rev}}$ 4 | Shows that the declared semantic strata are not only toy labels                     |
| Mutation-phase aggregate | The semantic-mutation track exceeds the closest baseline by +0.101 on mean score (Holm-adjusted $p = 0.046$ , Cliff's $\delta = +0.247$ ); the remaining pairwise contrasts are not significant                                                                        | Supports a modest, qualified mutation-score advantage rather than a dominance claim |
| Real-defect face         | The semantic-mutation track detects all 34 tabled real defects; the two generic baseline groups have nonzero detection in 7/34 and 8/34 cases, and the two ablated variants miss 19/34 and 17/34 cases, respectively, with 11 shared misses                            | Separates real-defect detection from aggregate mutant kill-rate                     |
| Non-nesting cases        | Four verified non-nesting defects (A-LAPACK-004, A-OPENBLAS-001, B-POCKETFFT-002, and E-ORDINARYDIFFEQ-001) spanning linear algebra, FFT, and ordinary-differential-equation solvers                                                                                   | Supports the semantic-effect fiber view rather than a single scalar hierarchy       |

This summary is intentionally a result table, not a corpus or benchmark description. The main manuscript uses it only to support construct separation: a relation family can score similarly on aggregate mutant kill-rate while differing sharply on real-defect detection, and different relation families can observe non-nested semantic-effect fibers.

*Count caliber.* The count of 34 is the frozen pre-registered mutation dataset (`data/mutation/results.jsonl`, transcribed in `data/mutation/REALDEFECT-FACE.md`), and its stratum composition above is the join of that dataset with the ledger meta-patterns. This is a different caliber from the defect ledger's `verified_full` census, which stands at 35 after E-PETSC-004 was upgraded from `candidate_full` on 2026-07-06 once its upstream fix was merged (ledger  $m_{\text{inv}}$  6 to 7, through family `f_inv.eqv` 4 to 5). The frozen dataset is not identical to the `verified_full` set: it omits A-MAGMA-002 ( $m_{\text{inv}}$ , a CUDA-GPU platform-gated case) and F-EIGEN-001 ( $m_{\text{conv}}$ , a sparse-storage representation case), and it retains C-GSL-001 ( $m_{\text{mono}}$ ), a `candidate_full` case carried with a disclosed reference-fix branch as its fixed arm. Accordingly the dataset  $m_{\text{inv}}$  6 equals the ledger  $m_{\text{inv}}$  7 less the platform-gated A-MAGMA-002.

*Fixed-arm provenance.* E-PETSC-004's upstream fix (merge request !9403, "Fix MatGetOrdering() on MPIAIJ subclasses") was merged on 2026-07-03. A 2026-07-06 re-fetch of the merged raw diff is byte-identical to the diff on which our fixed branch is built (`scripts/mutation/e-petsc-004/mr9403.diff`, 437 lines, no difference under `diff -q`); the fixed arm is a faithful back-port of that merged diff onto the v3.21.3 build, a same-base pairing in the Defects4J style rather than a build of merged main, and this is disclosed in the threats to validity. With this last case resolved, the `verified_full` cases in the 34-case industrial arm rest on merged upstream fixes or same-base back-ports of them, and the single `candidate_full` case in the set, C-GSL-001, rests on a disclosed reference-fix equivalent; the other 15 `candidate_full` cases, which still lack a merged upstream fix, lie outside the 34-case mutation set.

*Axis correspondence.* The industrial arm's five strata  $m_{\text{inv}}$ ,  $m_{\text{mono}}$ ,  $m_{\text{conv}}$ ,  $m_{\text{adj}}$ , and  $m_{\text{rev}}$  overlap the main experiment's five axes (inv, mono, conv, dyn, cmp) on three, namely inv, mono, and conv. The strata  $m_{\text{adj}}$  and  $m_{\text{rev}}$  instead exercise the NOETHER extension slots (adj, rev), whereas dyn and cmp have no industrial-arm representative. Inside

the dataset the family names carry the `f_` prefix (`f_mono.stat`, `f_conv.rate`, and so on); `f_rev.traj` is the time-reversal family under the dataset's own naming.

## References

- [1] Yue Jia and Mark Harman. 2011. An Analysis and Survey of the Development of Mutation Testing. *IEEE Transactions on Software Engineering* 37, 5 (2011), 649–678. doi:10.1109/TSE.2010.62
- [2] William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery. 2007. *Numerical Recipes: The Art of Scientific Computing* (3 ed.). Cambridge University Press.